

STALKER

Stock Market Analyzer & Alpha Engine v3.0

Complete System Documentation

Data Analysis Pipeline | Scoring Models | Risk Management

Ye document poora system explain karta hai ? data kahan se aata hai, kaise analyze hota hai, kaise shares pick hote hain, aur final picks tak kaise pahunchte hain.

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1. System Overview

STALKER ek automated stock screening system hai jo:

```

195+ Indian Stocks (NSE + BSE)
?
DATA FETCH (Yahoo Finance + NSE India API)
?
STAGE 1: Safety Filters ? Weak/Risky stocks OUT
?
STAGE 2: Market Environment Check (Bull/Bear/Neutral)
?
STAGE 3: 8 Alpha Scoring Models ? Har stock ko 0-100 score
?
PENALTY ENGINE ? Score se points kato (debt, volatility)
?
CONFIRMATION GATE ? BUY ya WATCH decide karo
?
STAGE 4: Portfolio Assembly ? Sector diversification + Correlation control
?
TOP 15 PICKS ? Email + Dashboard

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Files involved:

File	Kya Karta Hai
main.py	Scheduler ? kab kya chalega
screener.py	Core brain ? screening + scoring + portfolio
indicators.py	Technical indicators calculate karta hai
market_structure.py	Trend detect karta hai (uptrend/downtrend/sideways)
fundamentals.py	Company health score (debt, ROE, growth)
risk_manager.py	Stop loss, targets, position sizing, trade type
data_fetcher.py	Yahoo Finance se data lata hai + news analysis
config.py	Saari settings ek jagah

db_manager.py

MongoDB mein data store karta hai

2. Daily Schedule

Time	Task	Kya Hota Hai
7:00 AM	Pre-Market Analysis	Poore 195+ stocks scan, scoring, picks finalize. M
8:15 AM	Price Verification	Har pick ka price double-check. 2%+ drift ho toh p
8:30 AM	Morning Email	Top 15 picks ki email sabko bhejo (Brevo API)
9:20 AM	Record Open Prices	Market khulne ke baad actual open prices record ka
3:35 PM	Record Close Prices	Market band hone ke baad close prices record karo
4:00 PM	EOD Report	Profit/Loss calculate, performance email, mistakes

Weekend/Holiday pe kuch nahi chalta ? is_nse_holiday() function check karta hai.

3. Data Sources

3.1 Price Data (OHLCV)

- Source: Yahoo Finance (yfinance library)
- Period: Last 3 months daily candles
- Method: Bulk download 15 stocks at a time (yf.download)
- Columns: Open, High, Low, Close, Volume

3.2 Fundamental Data

- Source: Yahoo Finance ticker.info
- What: Market Cap, P/E, Debt/Equity, ROE, Revenue Growth, Profit Growth, EPS, Beta
- Cache: Local JSON file (data/fundamentals_cache.json) ? refreshes every 7 days

3.3 News & Catalysts

- Source: Yahoo Finance ticker.news
- What: Recent headlines ? keyword analysis ? sentiment (bullish/neutral/bearish)
- Cache: Local JSON file (data/news_cache.json) ? refreshes every 4 hours
- Keywords checked:

- Bullish: profit, growth, beat, upgrade, buy, outperform, expansion, contract win, acquisition, dividend, bonus, rally
- Bearish: loss, decline, downgrade, sell, underperform, weak, fraud, penalty, investigation, drop, miss, debt, default, crisis, layoff, crash

3.4 Market Indices

- NIFTY 50, Bank Nifty, Nifty IT, Nifty Pharma, Nifty Auto, Nifty FMCG, Nifty Energy, Nifty Metal
- Used for: Market regime (Bull/Bear/Neutral) + Sector trend detection

3.5 Live Prices (NSE India API)

- Direct NSE India API call for real-time prices during market hours
- Used in EOD report for accurate close prices

3.6 Rate Limiting Protection

- Agar Yahoo Finance 429 error de toh 5-minute cooldown activate hota hai
- Cooldown ke dauran saare network calls skip hote hain, cached data use hota hai
- Custom log filter automatically detect karta hai rate limit errors

4. Stage 1 ? Safety Filters

Purpose: Pehle galat, risky, ya illiquid stocks hatao. Ye stocks kabhi pick nahi honge.

Filter 1: Minimum Data (Hard)

Stock ka history < 20 daily candles ? REJECTED

Filter 2: Liquidity Check (Hard)

Average Volume (20-day) >= 1,00,000 shares ? PASS
 Average Traded Value >= ?2,00,00,000 (?2 Cr) ? PASS
 Dono conditions meet nahi ? REJECTED

Filter 3: Risk Profile (Hard)

Risk Score = (ATR% ? 150) + (Max Drawdown ? 15)

Risk Score > 5.0 (Bull market) ? REJECTED

Risk Score > 4.2 (Neutral/Bear) ? REJECTED

Risk Score Examples:

ATR%	Max Drawdown	Risk Score	Result
2%	10%	4.5	[OK] Pass (Bull) / [X] Fail (Neutral)
3%	15%	6.75	[X] Rejected
1.5%	8%	3.45	[OK] Pass

Filter 4: Data Quality (Hard)

Score starts at 100

- Missing Revenue Growth ? -10
- Missing ROE ? -10
- Missing Debt Metrics ? -10
- Missing Price/Volume Data ? -20
- Data older than 7 days ? -10

Score < 70 ? REJECTED

Filter 5: Market Structure (Hard)

Structure = "downtrend" (Lower Highs + Lower Lows confirmed) ? REJECTED

Downtrend ka matlab hai stock already gir rha hai.

Downtrend mein buy karna = loss.

Downtrend kaise detect hota hai:

- Last 4 swing highs mein agar koi lower hai (1%+ move down) ? Lower High count +1
- Last 4 swing lows mein agar koi lower hai ? Lower Low count +1
- Dono >= 1 ? DOWNTREND confirmed

5. Stage 2 ? Market Environment

5.1 Market Regime (NIFTY 50 based)

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NIFTY Close > 50 EMA AND Close > 200 MA ? BULL
NIFTY Close < 200 MA ? BEAR
Else ? NEUTRAL

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5.2 Sector Trends (7 sectors)

Each sector index (Bank Nifty, Nifty IT, etc.) ka trend check:

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Price > SMA10 > SMA20 ? "bullish"
Price < SMA10 < SMA20 ? "bearish"
Else ? "sideways"

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5.3 Strict Bull-Only Mode

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STRICT_BULL_ONLY_BUY = True (currently active)
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Agar market Bull nahi hai ? saare stocks WATCH mein jayenge, koi BUY nahi

6. Stage 3 ? Alpha Scoring Engine

Purpose: Har surviving stock ko 8 alag models se score karo (0-100 each),

phir weighted average nikalo ? Composite Alpha Score.

8 Scoring Models & Weights

#	Model	Weight	Score Range	Kya Measure Karta Hai
1	Relative Strength	20%	0-100	Stock ne Nifty ko kitna outperform kiya (percentil
2	Institutional Score	20%	0-100	Volume surge, green volume dominance, accumulation
3	Market Structure	15%	0-100	Uptrend/Breakout/Sideways structure
4	Sector Momentum	12%	0-100	Stock ka sector bullish/bearish/sideways hai
5	Technical Score	12%	0-100	RSI, EMA alignment, VWAP, MACD, Bollinger Bands

6	Fundamental Score	8%	0-100	Market cap, debt, ROE, growth, earnings
7	Earnings Catalyst	8%	0-100	Earnings surprise, profit growth
8	Opportunity Score	5%	0-100	Breakout proximity, support level, OHL pattern, 52

Alpha Score = (0.20 ? RS) + (0.20 ? Institutional) + (0.15 ? Structure) +
 (0.12 ? Sector) + (0.12 ? Technical) + (0.08 ? Fundamental) +
 (0.08 ? Earnings) + (0.05 ? Opportunity)

Model 1: Relative Strength Score (20%)

Kya karta hai: Stock ka 20-day return vs NIFTY 50 ka 20-day return

RS = Stock Return(20d) - NIFTY Return(20d)

Phir saare surviving stocks ke RS values ka percentile rank nikalta hai.
 Top performer = 100, Bottom = 0

Model 2: Institutional Score (20%)

Kya karta hai: Detect karta hai ki institutional buyers actively accumulate kar rhe hain ya nahi.

3 sub-components:

Sub-Score	Max Points	Logic
Volume Surge	40	Volume ratio $\geq 1.3x$? points. Below $1.3x$? 0
Accumulation Dominance	30	Green candle volume > 55% of total ? points
Momentum Days	30	Green days > 60% in last 15 days ? points

Volume ratio $1.0x$ (average) ? 0 points (no institutional activity)
 Volume ratio $2.0x$? 26.7 points
 Volume ratio $3.0x$? 40 points (max)

Model 3: Market Structure Score (15%) ? NEW

Kya karta hai: Stock ka overall trend pattern kya hai

Structure	Base Score	Description
Breakout	100	Price broke above resistance with volume

Uptrend	80	Higher Highs + Higher Lows
Sideways	30	Range-bound, no clear direction
Downtrend	0	Lower Highs + Lower Lows (already filtered in Stag

Bonus: Agar uptrend/breakout ka strength ≥ 75 ? extra +15 points

Structure kaise detect hota hai (market_structure.py):

1. Swing highs dhundho (last 8 candles left + 8 right ka peak)
2. Swing lows dhundho (same window ka trough)
3. Last 4 swing highs compare karo:
 - Har consecutive pair mein 1%+ upar gaya ? Higher High (+1)
 - 1%+ neeche gaya ? Lower High (+1)
4. Last 4 swing lows compare karo:
 - Higher Low count aur Lower Low count nikalo
5. Classification:
 - HH ≥ 1 AND HL ≥ 1 ? Uptrend
 - LH ≥ 1 AND LL ≥ 1 ? Downtrend
 - Else ? Sideways
6. Sideways mein agar today's close > 20-day range high (0.5%+ above) AND volume > 1.5x average ?
Breakout

Model 4: Sector Momentum Score (12%)

Sector Trend	Score
Bullish	100
Sideways	70
Bearish	20
Unknown	50

Sector trend detect hota hai sector index (Bank Nifty, Nifty IT, etc.) ke SMA10 vs SMA20 se.

Model 5: Technical Score (12%)

5 sub-components:

Sub-Score	Max Points	Logic
RSI Sweet Spot	30	RSI 55-70 ? 30pts. RSI 50-55 ? 15pts. RSI 70-75 ?
EMA Alignment	25	20 EMA > 50 EMA ? bullish structure confirmed

VWAP & EMA Slope	20	Price above VWAP ? 10pts. EMA slope upward ? 10pts
MACD Confirmation	15	MACD line > Signal AND histogram > 0 ? 15pts. Hist
Bollinger Squeeze	10	BB width < 70% of average ? low volatility ? break

VWAP kaise calculate hota hai:

$$\text{VWAP} = (\text{High} + \text{Low} + \text{Close}) / 3 \quad (\text{daily typical price})$$
 Agar close > VWAP ? intraday momentum bullish

RSI kaise calculate hota hai:

$$\text{RSI} = 100 - (100 / (1 + \text{RS}))$$

$$\text{RS} = \text{Average Gain} / \text{Average Loss} \quad (14\text{-day exponential})$$

 RSI 55-70 = healthy momentum zone
 RSI 75+ = overbought, reverse hone wala hai ? PENALTY
 RSI < 45 = weakness

Model 6: Fundamental Score (8%)

Score out of 10, then scaled to 0-100:

Metric	Max Points	Logic
Market Cap	2.0	$\geq \text{₹}5,000 \text{ Cr} ? 2.0. \geq \text{₹}1,000 \text{ Cr} ? 1.0$
Debt/Equity	2.5	$\leq 0.3 ? 2.5. \leq 0.8 ? 2.0. \leq 1.5 ? 1.0. > 1.5 ?$
ROE	2.0	$\geq 20\% ? 2.0. \geq 12\% ? 1.5. \geq 0\% ? 0.5. < 0\% ? 0$
Profit Growth	2.0	$\geq 20\% ? 2.0. \geq 10\% ? 1.0. < 0\% ? 0$
Earnings Beat	1.5	Surprise > 5% ? 1.5. > 0% ? 0.5. < -5% ? -0.5
P/E Ratio	0.5	$< 15 ? 0.5 \text{ (value stock)}. > 80 ? \text{alert}$

Price Filter (Hard): Price < ₹50 ya > ₹10,000 ? disqualified

Model 7: Earnings Catalyst Score (8%)

Condition	Score
Earnings Surprise > 10%	100
Earnings Surprise > 0%	80
Earnings Surprise < -5%	20
Profit Growth $\geq 20\%$ (fallback)	90

Profit Growth >= 0%	70
Profit Growth < 0%	30
No data	50

Model 8: Opportunity Score (5%)

Condition	Points
Price within 3% of resistance (breakout zone)	+35
Price within 3% of support (safe entry)	+25
OHL Bullish (Open = Low, buyers from start)	+25
Within 5% of 52-week high	+15
None of the above	0

7. Penalty Engine

Alpha Score ke baad penalties apply hoti hain:

Condition	Penalty
No recent earnings data available	-5
Debt/Equity > 1.2	-10
ATR/Price > 6% (extreme volatility)	-10
Bear market + Non-defensive sector	-15

Adjusted Alpha = Alpha Score + Penalties (max -30)

8. Confirmation Gate ? BUY vs WATCH

Sirf high alpha score se BUY nahi milta. 4 conditions ALL satisfy honi chahiye.

Buy Threshold (Regime-based):

Market Regime	Minimum Alpha Score
Bull	70

Neutral	74
Bear	80

4 Confirmation Conditions (ALL required for BUY):

#	Condition	Kya Check Hota Hai
1	Alpha Score $\geq$ Threshold	Score itna high hai ki worth buying hai
2	News Catalyst Exists	Sentiment bullish, ya catalysts found, ya news cou
3	Buyer Volume Confirmed	Volume ratio $\geq$ 1.5x OR Institutional Score $\geq$ 70
4	Strong Structure	Market structure = uptrend/breakout AND Technical

Charon meet ? BUY (with Stop Loss, Target 1, Target 2)
 Koi bhi fail ? WATCH (no stop loss, no targets ? sirf monitor karo)

Strict Bull-Only Override:

STRICT_BULL_ONLY_BUY = True
 Market != Bull ? FORCED WATCH (no matter what score hai)

9. Stage 4 ? Portfolio Assembly

Top 15 picks select karne ke liye 2 additional controls:

9.1 Sector Concentration Limit

Maximum 2 stocks per sector

Example: Agar 2 Banking stocks already selected, 3rd Banking stock skip hoga
 chahe uska score 95 ho.

9.2 Pearson Correlation Control

Agar 2 stocks ka price movement > 80% correlated hai (over 20 and 60 days)
 ? duplicate exposure hai, 2nd stock skip karo

Example: TCS aur Infosys typically 85%+ correlated hain.
 Agar TCS already selected hai, Infosys skip hoga.

Final Portfolio:

Candidates sorted by Adjusted Alpha Score (highest first)
 ?
 Sector limit check (max 2 per sector)
 ?
 Correlation check (skip if $r > 0.80$ with any existing pick)
 ?
 Top 15 stocks = FINAL PORTFOLIO

10. Stop Loss & Targets

Stop Loss Calculation (Only for BUY signals):

ATR Stop = Current Price - (2.0 ? ATR)
 Swing Stop = Swing Support ? 0.99 (1% below recent swing low)
 Final Stop = WIDER of the two (to survive daily noise)
 Hard Floor = Never more than 10% below entry

Example:

Entry = ₹500, ATR = ₹12, Swing Support = ₹480
 ATR Stop = $500 - (2.0 \times 12) = ₹476$
 Swing Stop = $480 \times 0.99 = ₹475.20$
 Final Stop = $\min(476, 475.20) = ₹475.20$ (wider stop)
 Hard Floor = $500 \times 0.90 = ₹450$ (won't go below this)

Target Calculation:

Risk = Entry - Stop Loss

Target 1 (Conservative) = Entry + Risk * 1.5 (1:1.5 R:R)

Target 2 (Ideal) = Entry + Risk * 2.0 (1:2 R:R)

Target 3 (Stretch) = Entry + Risk * 3.0 (1:3 R:R)

Example:

Entry = ₹500, Stop Loss = ₹475.20, Risk = ₹24.80

Target 1 = 500 + 24.80 * 1.5 = ₹537.20

Target 2 = 500 + 24.80 * 2.0 = ₹549.60

Target 3 = 500 + 24.80 * 3.0 = ₹574.40

11. Trade Types

Trade Type	Condition	Execution Rule
Gap & Go	Gap up >= 1.5% + uptrend/breakout	Buy if price stays above 9:15 AM Open in first 5 m
Breakout Trade	Structure = breakout	Buy if price breaks VWAP on 15m chart with 1.8x vo
Trend Continuation	Uptrend + RSI 45-58 (pullback zone)	Buy near 20 EMA on 15m chart when green support bo
Momentum Trade	Uptrend + RSI > 58	Buy if today's Open = Low (OHL Long setup)
Watchlist Only	Everything else	DO NOT trade. Monitor only

12. Price Verification (8:15 AM)

Morning email bhejne se 15 min pehle (8:15 AM pe), system har pick ka price double-check karta hai:

1. Yahoo Finance se live price fetch karo
2. 7 AM scan price se compare karo
3. Agar drift > 2% ? price update karo + stop loss/targets recalculate
4. Verification report generate karo (PASS/FAIL)
5. Updated data MongoDB + JSON mein save karo

13. EOD Report (4:00 PM)

Din ka result calculate karta hai:

1. Har pick ka actual open price aur close price fetch karo
2. P&L calculate: $(\text{Close} - \text{Open}) / \text{Open} \times 100$
3. VWAP trend check: $\text{Close} > \text{VWAP} ? \text{BULLISH}$
4. Nifty-relative strength: $\text{Stock return} - \text{Nifty return}$
5. Momentum score (0-10):
 - $\text{Close} > \text{VWAP} : +2.5$
 - Beat Nifty: +2.5
 - Close in top 20% of day's range: +2.5
 - Volatility > 1%: +2.5
6. Trailing stop loss for high-beta stocks ($\text{beta} > 1.2$)
7. Sector-wise performance grouping
8. Email report bhejo
9. Private mistakes audit generate karo (admin only)

14. Technical Indicators ? Complete List

Indicator	Formula	Used Where
EMA 20	Exponential Moving Average (20-period)	EMA alignment check, slope direction
EMA 50	Exponential Moving Average (50-period)	EMA alignment check
RSI (14)	$100 - (100 / (1 + \text{avg_gain}/\text{avg_loss}))$	Technical score, trade type
VWAP	$(\text{High} + \text{Low} + \text{Close}) / 3$ per daily bar	Above VWAP check
ATR (14)	EMA of True Range	Stop loss, risk profile
MACD	$\text{EMA}_{12} - \text{EMA}_{26}$, Signal = EMA_9 of MACD	Technical score (bullish/bearish)
Bollinger Bands	$\text{SMA}_{20} \pm 2 \times \text{StdDev}$	Squeeze detection (breakout potential)
Volume Surge Ratio	$\text{Today Volume} / 20\text{-day Avg Volume}$	Institutional score, confirmation
OHL Pattern	$\text{Open} = \text{Low} ? \text{Bullish}$, $\text{Open} = \text{High} ? \text{Bearish}$	Opportunity score
Gap %	$(\text{Today Open} - \text{Yesterday Close}) / \text{Yesterday Close}$	Trade type classification
52-Week Distance	$(\text{Close} - 52\text{w High}) / 52\text{w High}$	Opportunity score
Relative Strength	$\text{Stock 20d Return} - \text{Nifty 20d Return}$	RS score model

15. Config Values ? System Ki Settings

Setting	Value	Meaning
Stock Universe	195+ stocks	NIFTY 50 + Midcaps + BSE select
History Period	3 months	Daily candles
EMA Short	20	Short-term trend
EMA Long	50	Medium-term trend
RSI Period	14	Standard RSI
RSI Min	45	Below = weakness
RSI Max	78	Above = overbought
Volume Surge Ratio	1.8x	Minimum for "volume surge" boolean
VWAP Tolerance	0.5%	Tolerance for "above VWAP"
Gap Up Threshold	1.5%	Minimum gap for Gap & Go trade
Swing Lookback	8 candles	Window for finding swing highs/lows
Min Swing Move	1%	Minimum move to count as swing
Stop Loss ATR Mult	2.0x	Stop = 2x ATR below entry
Min Risk/Reward	1:1.5	Minimum acceptable R:R
Max Capital Risk	2%	Max risk per trade
Daily Loss Limit	3%	Stop trading after this
Top Picks Count	15	Maximum picks per day
Sector Limit	2	Max stocks per sector
Correlation Limit	0.80	Max allowed price correlation
Max Stock Price	?10,000	Skip expensive stocks
Min Stock Price	?50	Skip penny stocks
Min Market Cap	?5,000 Cr	Skip micro-caps
Max Debt/Equity	1.5	Hard limit
Strict Bull Only	ON	Force WATCH in non-Bull markets

Complete Flow Diagram

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????????????????????????????
?   195+ STOCKS UNIVERSE   ?
?   (NIFTY50 + Midcap +    ?
?   BSE Blue Chips)       ?
????????????????????????????
?
????????????????????????????
?   DATA FETCH           ?
?   ? 3mo OHLCV (yfinance) ?
?   ? Fundamentals (cached) ?
?   ? News headlines       ?
?   ? Market indices       ?
????????????????????????????
?
????????????????????????????
?   STAGE 1: SAFETY FILTERS      ?
?                               ?
?   [X] History < 20 days ? OUT    ?
?   [X] Volume < 1L or Value < ?2Cr ? OUT ?
?   [X] Risk Score > 4.2/5.0 ? OUT    ?
?   [X] Data Quality < 70 ? OUT      ?
?   [X] Downtrend confirmed ? OUT     ?
?                               ?
?   ~60-80 stocks typically survive ?
????????????????????????????
?
????????????????????????????
?   STAGE 2: MARKET CONTEXT      ?
?                               ?
?   NIFTY ? Bull / Neutral / Bear   ?
?   7 Sectors ? bullish/sideways/bearish ?
????????????????????????????
?
????????????????????????????
?   STAGE 3: ALPHA SCORING        ?
?                               ?
?   8 Models ? Weighted Average     ?
?   = Composite Alpha Score (0-100) ?
?                               ?
?   - Penalties Applied              ?
?   = Adjusted Alpha Score           ?
????????????????????????????
?
????????????????????????????
?   CONFIRMATION GATE             ?
?                               ?

```



```
? Alpha >= Threshold?      ?/?  ?
? News Catalyst?           ?/?  ?
? Buyer Volume?            ?/?  ?
? Strong Structure?         ?/?  ?
? Bull Market? (strict mode) ?/?  ?
?
? ALL ? ? BUY (SL + Targets)      ?
? ANY ? ? WATCH (no SL/targets)  ?
????????????????????????????????????????
?
????????????????????????????????????????
?   STAGE 4: PORTFOLIO ASSEMBLY    ?
?
? Sort by Adjusted Alpha (high?low) ?
? Max 2 stocks per sector          ?
? Skip if correlation > 0.80       ?
? Take top 15                      ?
????????????????????????????????????????
?
????????????????????????????????
?   FINAL 15 TOP PICKS      ?
?
? ? MongoDB saved          ?
? ? JSON saved             ?
? ? Email dispatched       ?
? ? Dashboard updated      ?
????????????????????????????????
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Last Updated: June 3, 2026

STALKER Alpha Engine v3.0